

MICHIGAN DEPARTMENT OF ENVIRONMENTAL QUALITY

INTEROFFICE COMMUNICATION

TO: Grace Scott, Field Operations Section, Water Bureau

FROM: Sarah Holden, Surface Water Assessment Section, Water Bureau
Sarah Holden

DATE: June 7, 2007

SUBJECT: Cadillac Wastewater Treatment Plant (WWTP) - Local Limits Review

Pursuant to your request, we have developed theoretical water quality-based effluent limits (WQBELs) for the Cadillac WWTP. The facility is authorized to discharge up to 3.2 million gallons per day (5 cubic feet per second [cfs]) of treated municipal wastewater to the Clam River. The lowest monthly 95 percent exceedance, harmonic mean, and 90-day 10-year flows for the Clam River are 2.1, 13, 2.9 cfs, respectively. Theoretical WQBELs are presented in the table below.

Parameter	Daily Maximum		Monthly Average	
	µg/l	lbs/day	µg/l	lbs/day
Silver	22	0.59	1.3	0.035
Arsenic	680	18	170	4.4
Cadmium	36	0.96	8.3	0.22
Chromium	2900	77	210	5.5
Copper	74	2	25	0.67
Cyanide, free	44	1.2	5.8	0.15
Lead	1600	44	100	2.7
Nickel	1800	47	110	2.9
Zinc	850	23	470	13
Molybdenum	58000	1500	3500	94
Selenium	120	3.2	5.5	0.15
Toluene	2600	69	300	8
Ethylbenzene	320	8.5	20	0.53
Xylene	730	19	45	1.2
Chloroform	11000	290	700	19
Lindane	1.9	0.051	0.03	0.0008
Phenol	6800	180	500	13
Styrene	2900	77	130	3.5
1,2-Dichlorobenzene	240	6.4	14	0.38

Please contact me if you have any questions.

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cc: Mike Stiffler, Cadillac District Supervisor, WB
Brenda Sayles/Facility File, WB

WATER BUREAU

JUN 8 2007

FIELD OPERATIONS
DIVISION